Md. Mokarram Chowdhury

Ames, Iowa, USA

mokarram@iastate.edu — Google Scholar — LinkedIn

EDUCATION

Iowa State University, USA — Aug 2023 – Present
Ph.D. in Computer Science

North South University, Bangladesh — Feb 2018
B.S. in Computer Science and Engineering

RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Mechanistic Interpretability, Model Compression, Efficient ML Systems, Trustworthy AI.

PUBLICATIONS

- **Md. Mokarram Chowdhury**, Daniel Agyei Asante, Ernie Chang, Yang Li. "IMPACT: Importance-Aware Activation Space Reconstruction." *ACL 2026 (to appear).*
- Daniel Agyei Asante, **Md. Mokarram Chowdhury**, Yang Li. "Decomposed Trust: Privacy, Adversarial Robustness, Ethics, and Fairness in Low-Rank LLMs." *Findings of ACL 2026 (to appear).*
- **Md. Mokarram Chowdhury**, Ahsanur Rahman. "Predicting Places of Revelation of Quran's Verses." *ICCIT 2020.*
- **Md. Mokarram Chowdhury**, et al. "Audio Gadget Recommendation by Fuzzy Logic." *CSOC 2019.*

ONGOING RESEARCH

- **Mechanistic Interpretability:** Internal representations in LLMs.
- **Efficient LLM Systems:** Compression, quantization, and scalable inference.
- **Evaluation & Benchmarking:** Large-scale model evaluation.

EXPERIENCE

Teaching Assistant, Iowa State University — Aug 2023 – Present
- Led recitations and supported large undergraduate courses; coordinated grading and course logistics.
- Mentored students and provided technical guidance in algorithms and systems topics.

Research Assistant, North South University — Sep 2022 – Aug 2023
- Developed and evaluated machine learning and deep learning models, with emphasis on NLP pipelines.
- Conducted data preprocessing, representation learning, and model optimization.
- Applied interpretability techniques to analyze and validate model behavior.

Lab Instructor, North South University — Jan 2018 – Aug 2022
- Delivered hands-on instruction in programming and systems courses; supervised labs and assessments.

TECHNICAL SKILLS

Programming: Python, C/C++
Machine Learning: PyTorch, Transformers, HuggingFace ecosystem, deep learning, representation learning
LLM Systems: Model compression, quantization, distributed training (FSDP), inference optimization (vLLM), large-scale model serving
Evaluation & Experimentation: Benchmarking, reproducibility, large-scale experiment management
Systems & Cloud: Linux, AWS, SLURM, GPU computing, parallel and distributed systems
Tools: Git, Docker, Jupyter, VSCode, Overleaf

SERVICE & LEADERSHIP

- Teaching Assistant for large-scale CS courses; experience in grading, coordination, and student support.
- Experience mentoring students and assisting in structured academic environments.
- Participated in IEEE Authorship Workshop.

AWARDS

- Champion, NSU Innovation Challenge and Capstone Showcase (2017)
- Winner, Techfest IIT Bombay Bangladesh Round (2016–17)
- Champion, IEEE INSB Robotics Competition (2016)